

Estimation methods of inequality of the population incomes.

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Abstract – in the given article survey of state-of-the-art methods of measurements of differentiation and concentration of incomes of the population is made. Significances of circumscribed parameters in a temporary slit are indicated.

Transition from a planned economy to market model has called significant fluctuations in all economic structure of Russia. But growth of an inequality of incomes came into one of the most negative socioeconomic outcomes of conducted reforms. To the analysis of parameters of differentiation of incomes, and of the most importance - their dynamics is paid the big attention at development of social policy in state-of-the-art Russia.

The differentiation of incomes, as a rule, is considered on the size of per-capita money income

Money incomes of population (MI) include incomes of the persons occupied in enterprise activity, the paid wages of hired workers, social payments (pensions, allowances, grants, insurance indemnities and other payments), and incomes of the property by the way the deposit percentages, to valuable papers, dividends and other incomes.

Average Per-capita incomes (APCI) (in a month) are estimated by division of annual volume of money incomes on 12 and on a population count.

Estimated as average, APCI, it is very sensitive to magnification or a diminution of a share of high-profitable or low-profitable groups of the population. In statistics of the majority of developed countries for performance of a cumulative level of incomes it is analyzed not average, but their median level, i.e. a level, that divides population on two equal (by quantity) groups – those who receive less income than this level, and those, who have the higher incomes. In this case we count the income of the average person, instead of

average returns of the abstract person. Also, modal incomes level is used, that is most common.

In table 1 the data of State Committee of Statistics for APCI - median and modal is presented for the period starting from 1980 to 1990, i.e. a period when the differentiation of incomes has been less expressed.

TABLE 1
INCOMES OF THE POPULATION TO THE USSR (RBL. / MONTH)

The income	1980	1985	1990
Average	109,6	125,8	164,6
Median	101,9	116,6	158,0
Modal	89,0	89,1	133,8

From the table it is visible, that the average returns on absolute value exceed median and modal income. Modal incomes values are strongly attracted by lower levels of distribution and deviate from the median in the lower side. However whether a modal value is in this or that interval is quite random – it's enough to have a little change in distribution to move a modal value in the other interval.

Thus, use of a parameter of average returns results in essential overestimate of a level of incomes of the most part of the population that hides process of their differentiation

However, the official statistics grants the information only on per-capita money incomes. Dynamic of average per-capita money incomes is presented in Table 2.

With the purpose of study of differentiation the distribution of the population on a level of per-capita money income.

Simulation of the population on per-capita money income basis allows spreading outcomes of sampling observation to all population of Russia or separate subjects of the Russian Federation.

TABLE 2
AVERAGE PER-CAPITA MONEY INCOMES

Years	1992	1995	1998	1999	2000	2001	2002	2003
APCI the population, rbl. (till 1998 one thousand rbl.)	4,0	515,5	1013	1663	2288	3075	3964	5142
Real available MI, in percentage by last year	52	85	84	88	112	109	111	115

Reference: [1]

Distribution is calculated on the basis of balance of money incomes and costs of the population, provided by

the Central bank of Russia and APCI samples of domestic economy.

Federal service of state statistics of Russia is calculating the logarithmical normal model (LNM). However there are examples of other models use: the dynamic model based on the equation of the Fokker-Blanc, the "mix" of two beta-distributions, gamma-distribution, etc. In opinion of many contributors LNM results in the extremely deformed estimations and significant underestimating of estimations of an inequality and poverty.

In Russia the research of the optimal model for the mathematical description of distribution of the population

TABLE 3
DISTRIBUTION OF THE POPULATION ON AVERAGE PER-CAPITA MONEY INCOMES (IN %)

	1997	1998	1999	2000	2001	2002	2003
All population including with APCI rbl. per month (up to 1998) thousands rbl.	100	100	100	100	100	100	100
Up to 1000,0	67,0	62,7	33,6	20,1	12,4	6,6	3,3
1000,1-1500,1	174,4	19,6	24,3	20,7	14,9	10,4	6,5
1500,1-2000,0	7,8	8,9	16,0	17,0	14,3	11,7	8,5
2000,1-3000,0	5,5	6,3	15,8	21,1	21,7	20,9	17,8
3000,1-4000,0	1,5	1,7	5,9	10,2	13,5	15,2	15,2
4000,1-5000,0					8,2	10,5	11,8
5000,1-7000,0	0,8	0,8	4,4	10,9	8,2	12,0	15,4
From above 7000,0					6,8	12,7	21,5

Reference: [1]

Distribution of the population on a level of APCI represents indicators of a number (or a share) of the population which have been grouped together in given intervals.

For study of differentiation of incomes of the population on PCI the rearrangements of households are conducted:

- On decile groups - ten groups, 10% of households in each
- On quintile groups - five groups, 10% of population in each
- On a buying ability of the population - on groups, aliquot to value of a cost of living

The analysis of differentiation of incomes consists in use of the information on distribution of the population under incomes only within the framework of extreme groups (poor and affluent), and the analysis of concentration - generalizes all distribution of the population under incomes.

on the per-capita income in conditions of transitional and market economy is performed now.

In countries of Western and the Eastern Europe as the mathematical description of distribution on the incomes the "mix" of logarithmically normal distribution for the basic part of the population and Pareto-distribution for high-profitable groups of the population is used. [2]

Distribution of the population on APCI is presented in table 3.

Population incomes differentiation indicators: Funds coefficient, Decile differentiation coefficient.

Population incomes concentration indicators: Lorenz curve, Lorenz coefficient, Herfindale coefficient, Gini coefficient.

Funds coefficient is a parameter of excess of incomes of 10 % of the population having the highest returns, and 10 % of the population having the lowest incomes. It represents the correlation of average coefficient in the upper and lower decile borders and shows, in what times the income of 10 % of the most profitable population on the average is more than the income at 10 % of the population with the least incomes.

The funds coefficient is one of the most widespread factors in statistics of incomes of the population. The data on this factor is presented in table 4.

TABLE 4
FUND COEFFICIENT (IN TIMES) IN THE RUSSIAN FEDERATION IN 1990 - 2003

1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003
4,4	4,5	8,0	11,2	15,1	13,5	13,0	13,5	13,8	14,0	13,8	14,0	14,0	14,3

Reference: the Federal service of state statistics

For the **decile differentiation coefficient** the extreme (the first and the ninth) deciles are calculated. The decile here is meant as a variant that divides ranked by the incomes aggregate on ten equal by volume groups (D_i). Decile differentiation coefficient is calculated using the function (1) on the grouped data:

$$D_i = X_{D_{i\min}} + \frac{L(k_i \sum F - \text{cum} F_{D_{i-1}})}{F_{D_i}} \quad (1)$$

Where:

D_i - i-th decile

i – decile index

$i = 1 - 9$ (9 deciles are calculated)

$X_{D_i \min}$ – the lowest interval border, containing i -th decile (is determined by accumulated parts)

L – value of an interval on the income

K_i – coefficient, corresponding to the decile index: for $D_1 K_1 = 10$, for $D_2 K_2 = 30$, ..., and $D_9 K_9 = 90$

$\sum F$ – Volume of aggregate (an aggregate number of the population);

$cumF_{D_{i-1}}$ – Accumulated detail in the interval, preceding to an interval containing i -th decile;

F_{D_i} – frequency of an interval containing i -th decile.

First decile characterizes the maximal income of 10 % of the least profitable population. The ninth decile, located in a penultimate interval, characterizes the minimal income of 10 % of the most profitable population.

Decile coefficient of differentiation is calculated as a ratio D_9 / D_1 and shows in how much times the minimal income of 10% of the most profitable population exceeds the maximal income of 10% of the least profitable population.

The data on decile differentiation coefficient is presented in table 5.

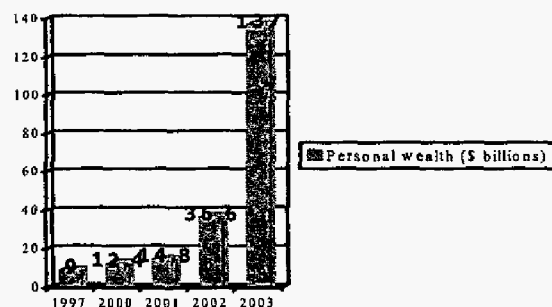
TABLE 5
DECILE DIFFERENTIATION COEFFICIENT IN THE RUSSIAN
FEDERATION IN 2001 - 2003

	2001	2002	2003
Decile coefficient (in times)	6,6	6,6	6,7

Reference: [1]

However, it is necessary to note, that for the last decade, in accordance with developing economy of the world, there is information on the businessmen possessing essential personal financial assets, exceeding million American dollars. These persons, as a rule, do not fall under diagnostic studies of the family budgets taken by statistical services and are not reflected in indicators of incomes differentiation of different groups of the population. According to the journal "Forbes" in Russia there are more billionaires in relation to gross national product, than in any other country of the world, - 36 persons on \$458 billions. Certainly, it is impossible to compare directly a wealth of the richest businessmen to

gross national product - first measures the stock, the second turnover, - however the fact remains the fact: the wealth of 36 richest people of Russia (\$110 billion) is equal to 24 % of gross national product of country. In February, 2004 when Forbes published the world rating of billionaires, the doubtless leader on number of rich men was the USA - 277 billionaires. However, the cumulative wealth of the American capitalists (\$651 billion) hardly holds out up to 6 % from gross national product - \$11 000 billion. On the second place there was Germany - 52 billionaires at gross national product in \$2100 bln. Japan, the second most powerful economy in the world, is characterized by the smaller concentration of the capital - only 22 owners of a milliard wealth and th gross national product of country in \$3700 bln.[3] Generalized data of the journal "Forbes" on the personal wealth of the richest Russian businessmen are shown on the figure 1.



1997 - 4 persons 2000 - 8 persons, 2001 - 8 persons,
2002 - 17 persons, 2003 - 100 persons

Figure 1: an estimation of the richest businessmen of Russia wealth

Reference: [3]

Defect of decile differentiation coefficient and funds coefficient is in partial use of information of the population distribution over the incomes only within the framework of the extreme decile groups. For studying of the incomes differentiation on all distribution the rearrangement on quintile groups, grouped by 20% of population, is performed. Quintile rearrangement of money incomes of Russian Federation population is presented in table 6.

TABLE 6
DISTRIBUTION OF TOTAL AMOUNT OF MONEY INCOMES (IN %) ON 20% GROUPS OF THE POPULATION

	1991	1992	1993	1994	1997	1999	2000	2001	2002
MI all, including	100	100	100	100	100	100	100	100	100
1	11,9	6,0	5,8	5,3	6,0	6,1	6,1	5,6	5,6
2	15,8	11,6	11,1	10,2	10,2	10,4	10,6	10,4	10,4
3	18,8	17,6	16,7	15,2	14,8	14,8	14,9	15,4	15,4
4	22,8	26,6	24,8	23	21,6	21,1	21,2	22,8	22,8
5	30,7	38,3	41,6	46,3	47,4	47,6	47,2	45,8	45,8

Reference: the Federal service of state statistics

The differentiation indicators, summarizing all distribution of the population over the incomes, include incomes concentration coefficients of Lorenz and Gini. They are part of the system of estimations, known as the methodology of Pareto – Lorenz – Gini, widely used in the modern foreign social statistics.

Calculation of the mentioned coefficients is made on the basis of the distribution of households the average (per one) income data. The whole number of income recipients is divided into 5 equal quintile groups and the share of the income each group owns is calculated.

On the obtained accumulated summaries the **curve of Lorenz** is constructed. This curve is graphical representation of the incomes concentration level.

For the representation of the Lorenz curve on the system axes with the percentage scale from 0 to 100 the cumulative summaries of the distribution are put: on the horizontal axis – quintiles of the income recipients, on the vertical axis – quintiles of the obtained incomes. Lorenz curve is drawn by the junction of the points, corresponding to cumulative percentage shares of incomes, obtained by quintile groups of population. With the even distribution of the incomes, every 20% group of population would have the fifth part of the society income. On the graph it is represented by the diagonal of a square and is treated as a line of even distribution. With uneven distribution, the "concentration line" is downwards concaved curve. The more this line differs from the diagonal, the more irregularity of the incomes distribution is, its concentration is accordingly higher.

The Lorenz curve for the Russian Federation is presented in figure 2.

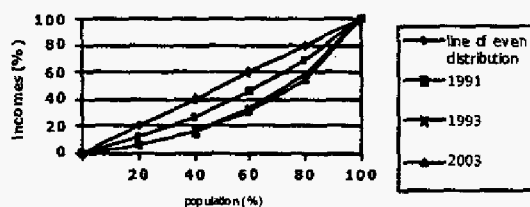


Figure 2. The Lorenz curve

Reference: this data is calculated by the author on the basis of the data of the Federal service of state statistics.

On the Lorenz curve basis the Lorenz Coefficient (L, function 2) and Herfindale coefficient (H, function 3) can be calculated:

$$L = \frac{\sum_{i=1}^n |w_i - d_i|}{2} \quad (2)$$

$$H = \sum_{i=1}^n (d_{y_i})^2 \quad (3)$$

Where

w_i – shares of units of population

d_{y_i} – shares of an investigated summarized indicator

n – number of social groups

However, the Lorenz coefficient is not used on practice in the population incomes differentiation analysis, and the Herfindale coefficient is used quite seldom.

As the other indicator of the level of differentiation of population over the incomes level, an index of incomes concentration, or Gini coefficient, is used. It represents the nature of distribution of the whole sum of population incomes between its (population) separate parts.

Gini coefficient is calculated as the ratio of square of the segment, which is formed by the line of actual distribution (Lorenz curve) and line of even distribution, to the square of the triangle that is lower than even distribution line.

Its value can vary from 0 up to 1. In case of the even distribution of incomes, the coefficient stays close to 0 values. The higher coefficient value is, the more social incomes are nonuniformly distributed. In the basis of Gini coefficient that is based on the Lorenz curve, the idea lays, that extreme positions in the distributions of the incomes or welfares are egalitarian (when all participants receive equal shares) or non-egalitarian (when the sole participant of distribution receives all welfares). The first case describes absolute equality, and the second – absolute inequality in distribution.

Mathematically, the GINI coefficient can be calculated using following function:

$$G = 1 - 2 \frac{\sum_{i=1}^n x_i \text{cum} y_i}{\sum_{i=1}^n x_i y_i} \quad (4)$$

Where

Y_i – the income share, that the i -th social population group achieves

X_i – the population share of i -th social group in an aggregate number of population

N – social groups quantity

$\text{cum} Y_i$ – cumulative income share

Also, Gini coefficient can be calculated on the average income basis ($\bar{y}$):

$$G = \frac{\sum_{i=1}^n \sum_{j=1}^n |y_i - y_j|}{2n^2 \bar{y}} \quad (5)$$

Where:

n – quantity of domestic economies

y_i, y_j – income of domestic economies

y – average households income

Gini coefficient is calculated with the purpose of the study of character of fluctuations in incomes distribution of society, and also for determination of inter-regional and international comparisons in a level of concentration of incomes. For today Gini coefficient is most "popular", both in foreign, and in domestic statistics, coefficient for the analysis of incomes distribution in society. The value of the Gini coefficient for the Russian Federation presented in table 7.

TABLE 7
GINI COEFFICIENT

1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003
0.260	0.289	0.398	0.409	0.381	0.375	0.381	0.398	0.399	0.399	0.398	0.398	0.400

Reference: the Federal service of state statistics

Besides the coefficients, listed above, for an estimation of inequality of incomes of the population the Atkinson index and entropy measures are used. These methods are used basically in the research jobs of Russian program of economic researches. Official statistics do not provide any information on these indicators.

Atkinson index

In the basis of Atkinson index lays the concept of equivalent income y_d , i.e. the least cumulative per-capita income that at income distribution would lead to the same value of social prosperity as with the existing incomes inequality. The public welfare, in turn, is determined as the sum of functions of individual functions of welfare. As individual functions of welfare depend only on the income, in empirical calculations the social inequality is reduced in essence to an inequality of incomes of the population.

Atkinson index is calculated using the following formula:

$$A_d = 1 - \left\{ \frac{1}{N} \sum_{i=1}^N \left[\frac{y_i}{y} \right]^{1-d} \right\}^{1/(1-d)} \quad (6)$$

Where

$0 < d < \infty$ - parameter, estimating the expected level of equality

$\bar{y}$ - per-capita income

y_i - i -th recipient income

N - total recipients quantity

The values of this coefficient vary from 0 up to 1. And 0 is indicating that there is no inequality.

The most disadvantage of Atkinson index is that it is very difficult to set value of d parameter. And there's no clear and furthermore formalized solution can be found. [4]

Entropy measures

As the inequality indicators in the public statistics the so-called entropy measures are also used. They are related to generalized class of entropy measures

In the general form they look like:

$$I_c(y) = \frac{1}{N(c^2 - c)} \sum_{i=1}^n \left[\left(\frac{y_i}{y} \right)^c - 1 \right] \quad (7)$$

Where

N - households quantity

Y_i - i -th domestic economy income

$\bar{y}$ - average income

$I_c(y)$ value can vary from 0 up to ∞ . And 0 will mean equal income distribution, and the higher this coefficient value is, the higher inequality level is.

Parameter c represents the weight, that is applied to the difference in the incomes of the different parts of income distribution, and can't take any real value. At small values of c , $I_c(y)$ is most sensitive to changes in the "lower tail" of the distribution, i.e. at the low incomes level. And at the higher values of c , $I_c(y)$ is most sensitive to the changes in the "upper tail" of income distribution. The most often-used are values 0, 1 and 2.

$I_c(y)$ with $c = 1$ - correlate with index, that was proposed by Till and is called Till measure:

$$I_1(y) = \frac{1}{N} \sum_{i=1}^N \frac{y_i}{y} \log \frac{y_i}{y} \quad (8)$$

Some authors consider that entropy measures have some advantage in comparison with other meters of inequality as can be decomposed into the components relating to different population groups, i.e. represent the weighted sum of indexes for the different households groups. [4]

Purchasing capacity population distribution

On the modern stage the most widely used for the estimation of the irregularity of incomes distribution is the population distribution over purchasing capacity. Purchasing capacity is determined as ratio of APCI to value of the cost of living (CL).

For this analysis the grouping into divisions with the same purchasing capacity is made. In the calculating of such grouping, the incomes intervals aliquot to CL value are used. In each segregated group are included households with the same purchasing capacity. On the basis of provided data the following coefficients are calculated:

contrast ratio on population quantity, as a ratio of the population shares in the extreme groups; contrast ration on per-capita incomes, as a ration of the income shares in the highest and lowest groups; general contrast ratio, equal to the product of the stated ratios.

The grade of purchasing capacity, per se, has comparative nature and provides comparison of values of the corresponding indicators in temporary (mainly retrospective) or spatial (cross-territorial or inter group) aspects.

Such comparisons are necessary now for the estimation of the effect of realized socio-economic transformations

on the society life. Comparison by various regions of country is needed for calculation of divergences in a level and conditions of living of the population between them and for the determining reasons analysis.

At realization of temporary, inter-regional and inter group comparisons of a standard of living of the population the following indicators are used general indicators of a variation and comparative indicators of a variation.

Information References for realization of the analysis of incomes differentiation of the population in modern Russia are:

- The Federal service of state statistics
- The World bank
- The Data of the Russian monitoring an economic situation and health of the population (RLMS)
- Alternative References (the private{individual} researches, given mass media, etc.)

It is necessary to note, that the information given by these References can significant differ from each other. So, for example, GINI coefficient calculated using RLMS data - in 1994 was 0,45, in 1999 - 0,47. And the same indicator, calculated using the Federal service of state statistics data was equal to 0,409 and 0,398 accordingly. By results of the calculations made by A. Suvorov and E. Ulyanova, the funds coefficient in Russia was 62 times in 1996 and 26 times in 1995 whereas the official statistics estimated it in 13,0 and 13,5 times accordingly. [5]

On the basis of the aforesaid we can make conclusions:

- 1) On the modern stage for economic-statistical research there is a plenty of the methods, those are used for estimation of population differentiation over money incomes.
- 2) All indicators of differentiation (both simple and more complicated) give quite a crude estimation of inequality, since the richest and poorest society members do not participate in sampling observation of the population.
- 3) However, the analysis of dynamics of differentiation indicators assumes revealing tendencies in process development and remains the necessary information for the prevention of negative effects, those lead to social tension forming.
- 4) The most actual are interregional and inter group estimations of population purchasing capacity

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